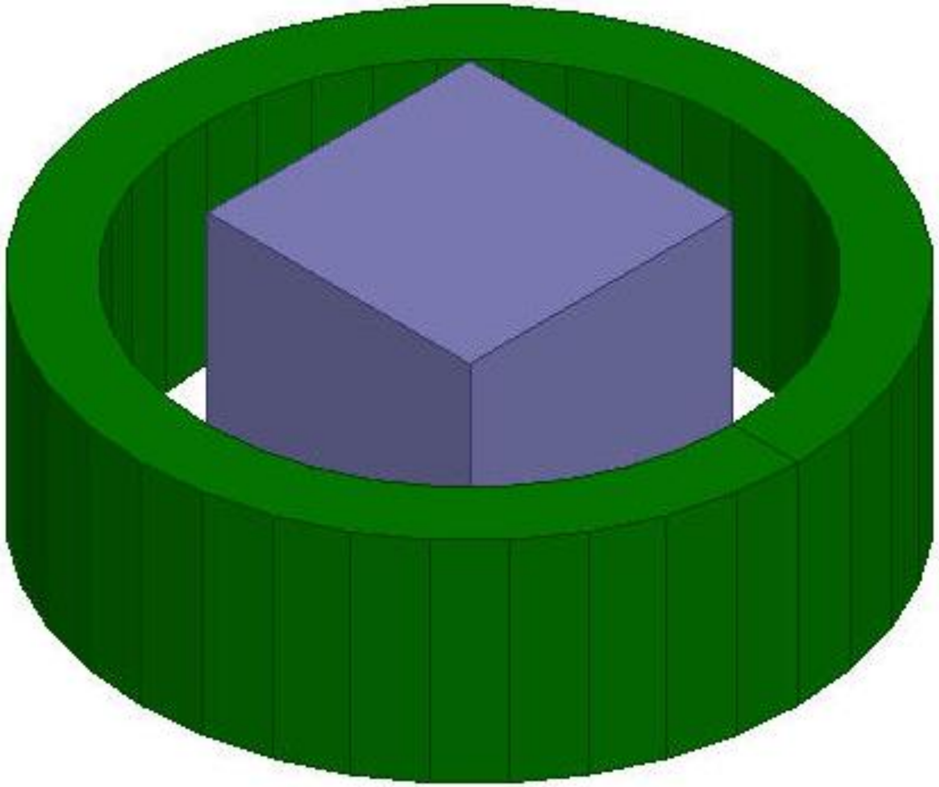
Workshop 4.2: Parametric Analysis

Release 2020R2

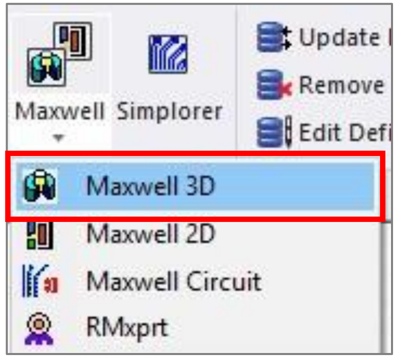
Overview

- **Parametric Analysis using a 3D model consisting of a Coil and a Slug**
  - **This workshop describes the steps required to setup a parametric analysis offered through Optimetrics.**
  - **A simple magnetostatic problem will be used to demonstrate the setup. The coil current and the dimensional length of an iron slug will be varied and the impact of changes in above parameters on the force exerted on the slug will be observed.**

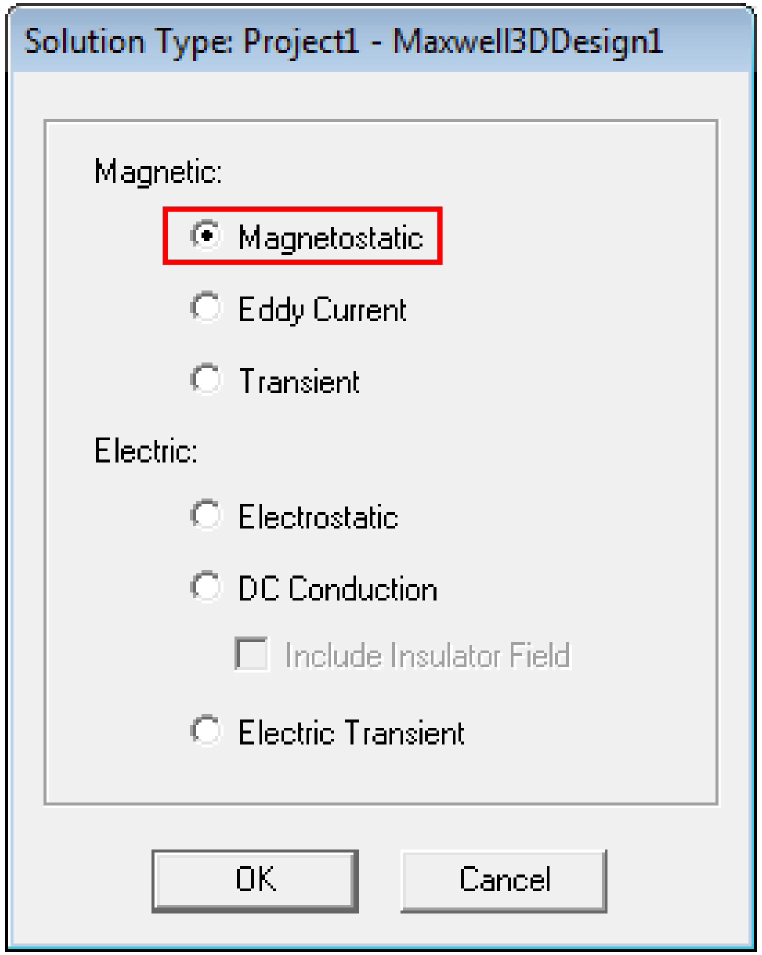


Problem Setup

- **Insert Design**
  - **Select the menu item *Project → Insert Maxwell 3D Design*, or click on the icon in drop down list Maxwell on panel Desktop**



- **Set Solution Type**
  - **Select the menu item *Maxwell 3D → Solution Type***
  - **Choose *Magnetic → Magnetostatic***
  - **Click the OK button**
- **Set Model Units**
  - **Select the menu item *Modeler → Units***
  - **In Set Modeler Units window,**
    - **Select units: mm (millimeters)**
    - **Press the OK button**

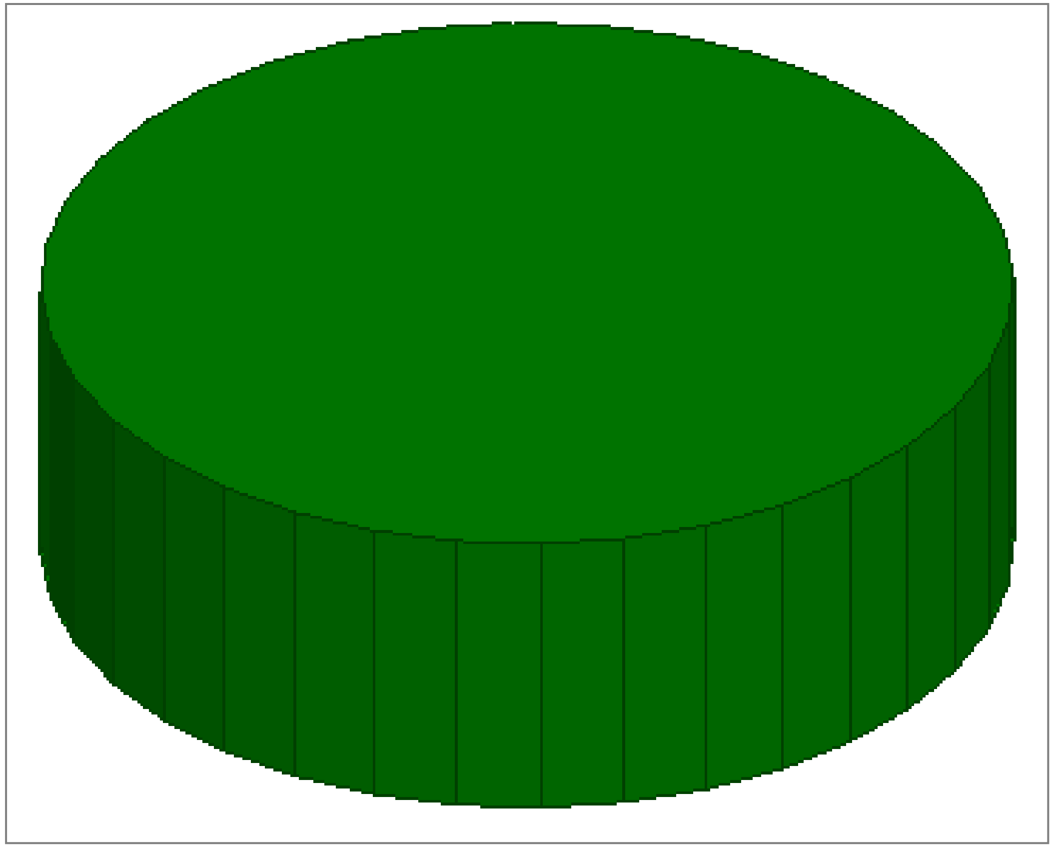


Create Coil

- Create Regular Polyhedron
  - Select the menu item *Draw → Regular Polyhedron*
    - Using the coordinate entry fields, enter the center of the base

      X: 0, Y: 0, Z: 0, Press the *Enter* key
    - Using the coordinate entry fields, enter the radius

      dX: 1.25, dY: 0, dZ:0.8, Press the *Enter* key
  - Number of Segments: 36
  - Change the name of the Object to Coil
  - Change the material of the object to Copper
  - Change the color if desired



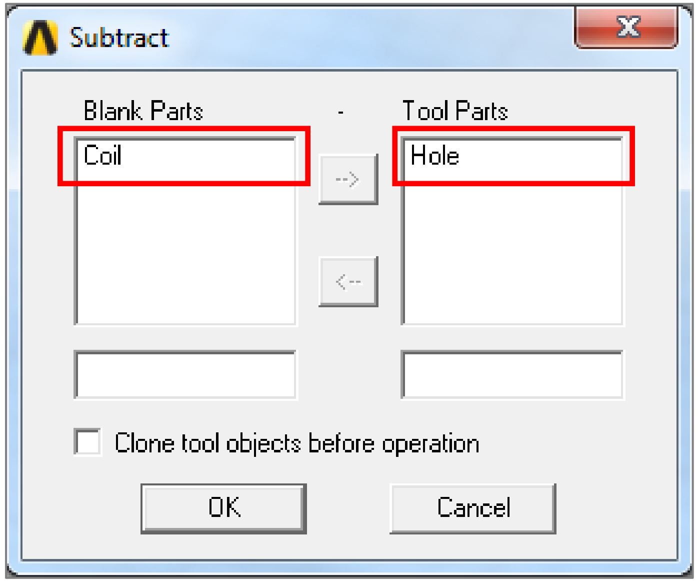
Create Coil

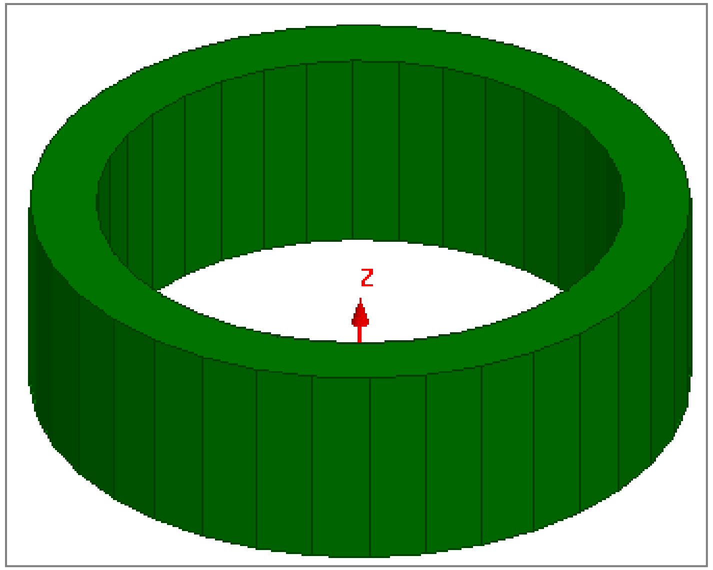
- Create Second Polyhedron
  - Select the menu item *Draw → Regular Polyhedron*
    - Using the coordinate entry fields, enter the center of the base

      X: 0, Y: 0, Z: 0, Press the *Enter* key
    - Using the coordinate entry fields, enter the radius

      dX: 1, dY: 0, dZ: 0.8, Press the *Enter* key
    - Number of Segments: 36
  - Change the name of the Object to Hole

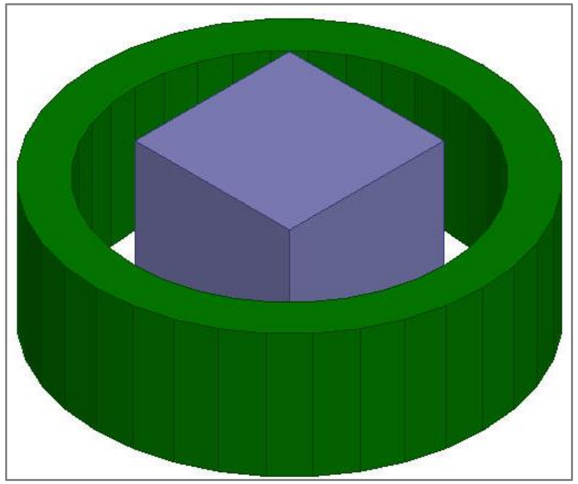
- Subtract Objects
  - Press Ctrl and select the objects Coil and Hole from the history tree
  - Select the menu item *Modeler → Boolean → Subtract*
    - Blank Parts: *Coil*
    - Tool Parts: *Hole*
    - Click the OK button



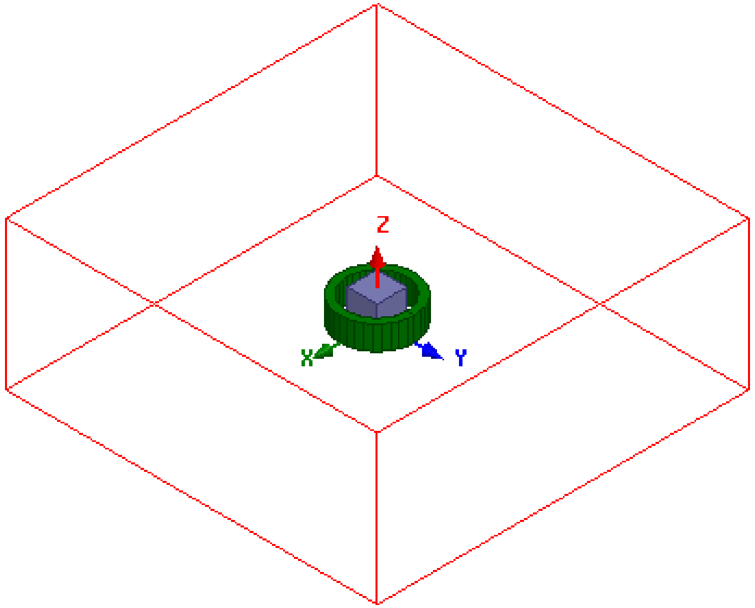


Create Slug

- Create Iron Slug
  - Select the menu item ***Draw → Box***
    - Using the coordinate entry fields, enter the box position
      X: -0.5, Y: -0.5, Z: 0, Press the ***Enter*** key
    - Using the coordinate entry fields, enter the opposite corner
      dX: 1, dY: 1, dZ: 1, Press the ***Enter*** key
  - Name the resulting object **Slug** and change material to **Iron**

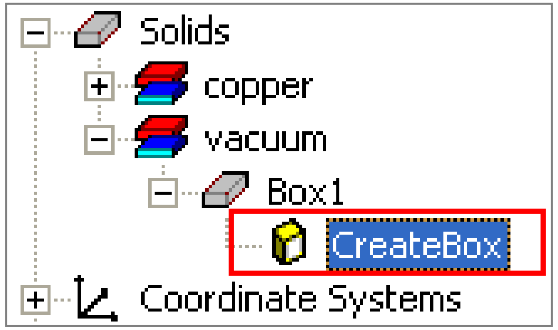


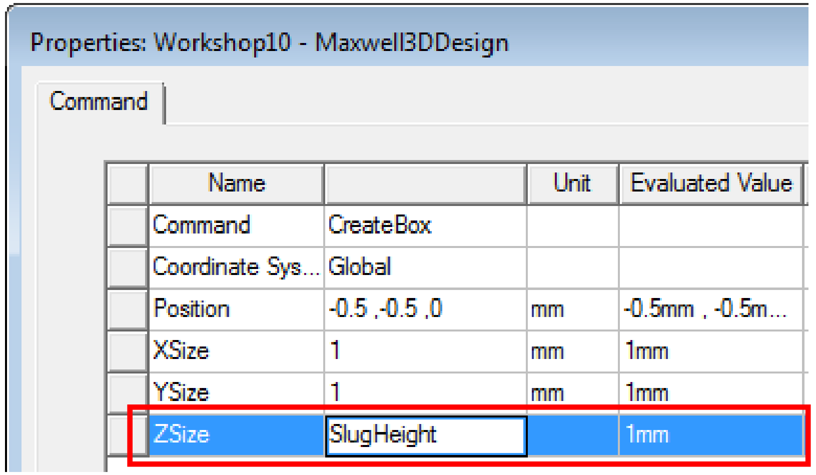
- Create Simulation Region
  - Select the menu item ***Draw → Region***
  - In Region window,
    - Padding all directions similarly: ☑ Checked
    - Percentage Offset:
    - Value: 200
    - Press OK

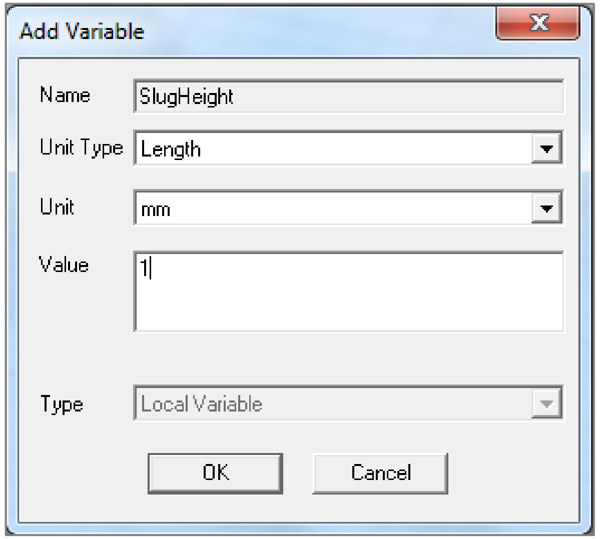


Add Parameter SlugHeight

- **Add parameter for Slug Height**
  - **Expand the history tree for the object Slug and double click on CreateBox**
  - **In Properties window,**
    - **For ZSize type SlugHeight, press Tab to accept**
    - **The pop-up Add Variable window appears**
    - **Unit Type: Length**
    - **Unit: mm**
    - **Value: 1**
    - **Press OK**







Note: By defining a variable name (SlugHeight) it becomes a design variable. All design variables are accessible by selecting menu item Maxwell 3D → Design Properties. If variable name is appended by symbol $, it will be defined as project variable and can be accessed across all the designs

Create Coil Terminal

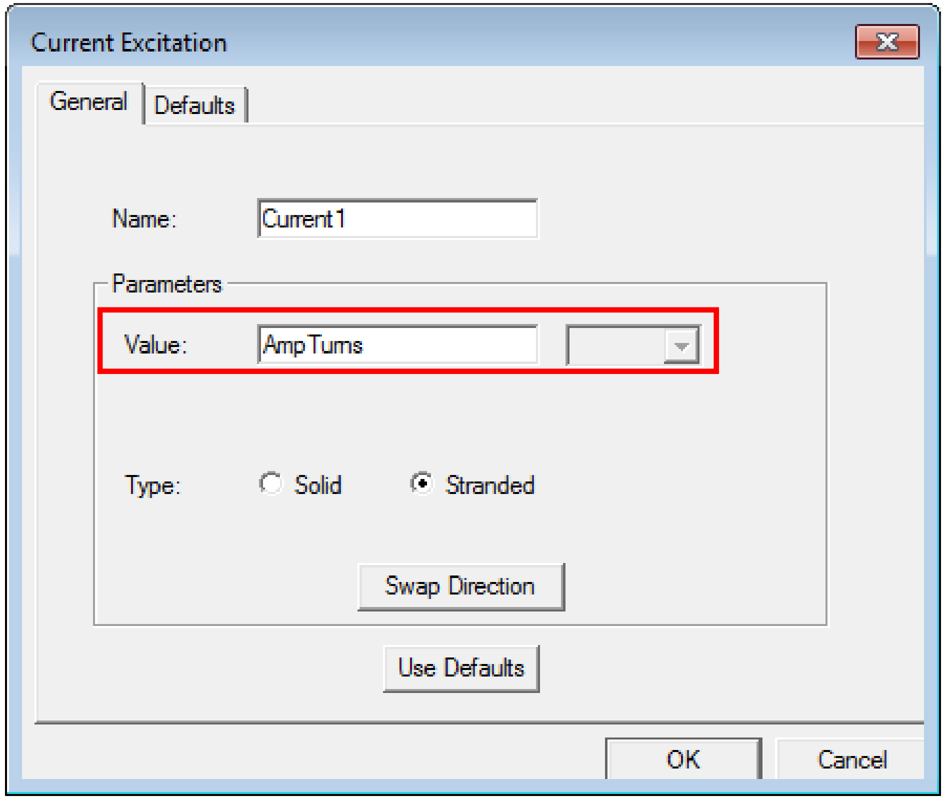
- Create Coil terminal
  - Select the object Coil from the history tree
  - Select the menu item ***Modeler → Surface → Section***
  - Section Plane: YZ
  - Press OK
  - Change the name of the resulting sheet to Terminal

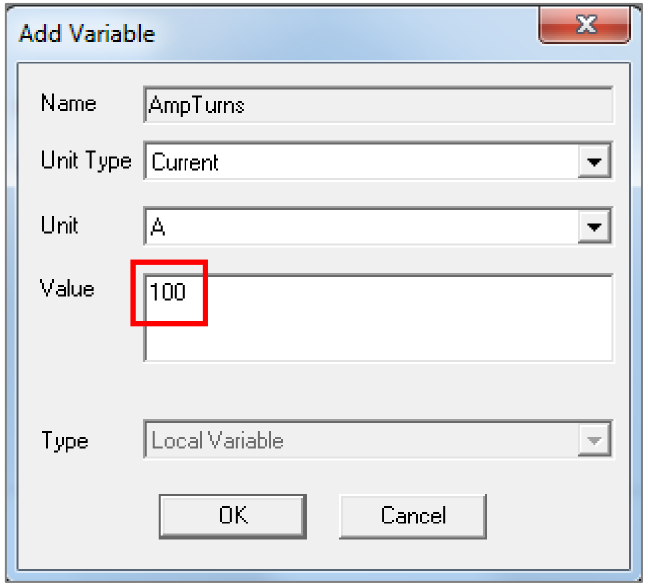
- Separate Sheets
  - Select the sheet Terminal from the history tree
  - Select the menu item ***Modeler → Boolean → Separate Bodies***

- Delete Extra Sheet
  - Select the sheet Terminal_Separate1 from the history tree
  - Select the menu item Edit → Delete

Assign Excitation

- Assign Excitation
  - Select the sheet Terminal from the history tree
  - Menu item *Maxwell 3D → Excitations → Assign → Current*
  - In Current Excitation window,
    - Name: Current1
    - Value: AmpTurns
    - Type: Stranded
    - Press OK
  - In Add Variable window,
    - Unit Type: Current
    - Unit: A
    - Value: 100
    - Press OK



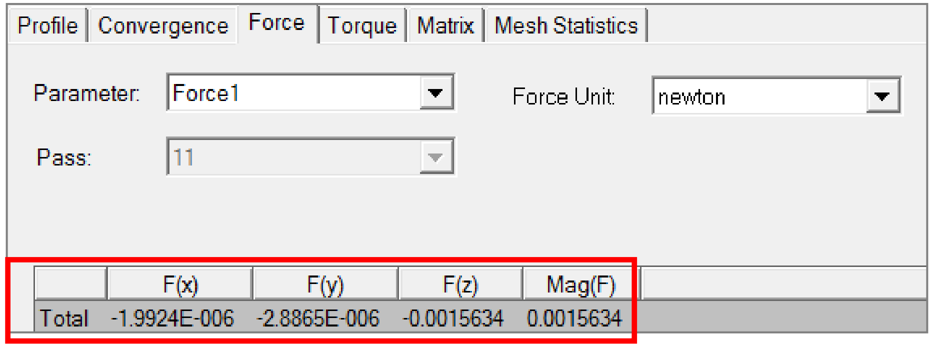


Analyze

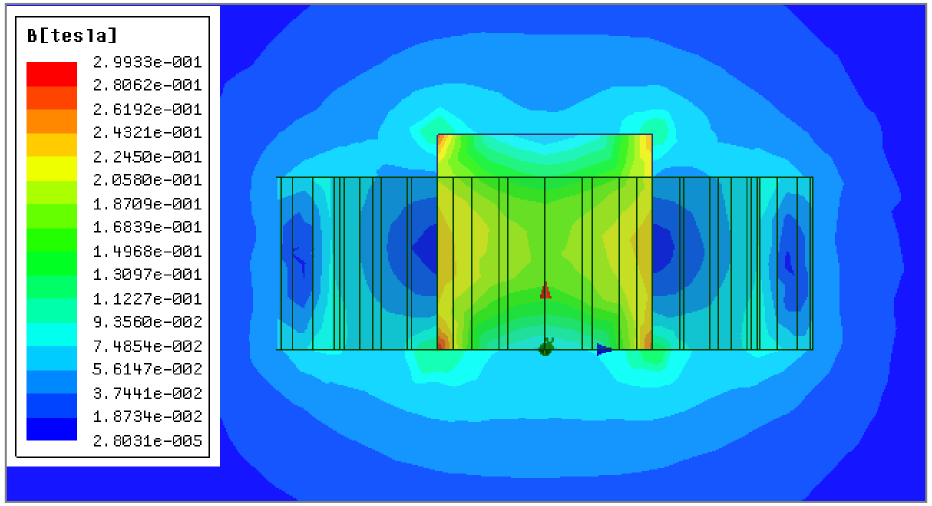
- Assign Force Calculation
  - Select the object Slug from the history tree
  - Menu item *Maxwell 3D → Parameters → Assign → Force*
  - In Torque window,
    - Name: Force1
    - Type: Virtual
    - Press OK
- Create an analysis setup:
  - In Project Manager window *RMB on Analysis → Add Solution Setup*
  - Solution Setup Window:
    - General tab
    - Maximum Number of Passes: 15
    - Click the OK button
- Run Nominal Solution
  - In Project Manager window *RMB on Setup 1 → Analyze*

Results

- View the Solution Data:
  - Menu item ***Maxwell 3D→ Results→ Solution Data***
    - To view Force values
    - Select the Force tab



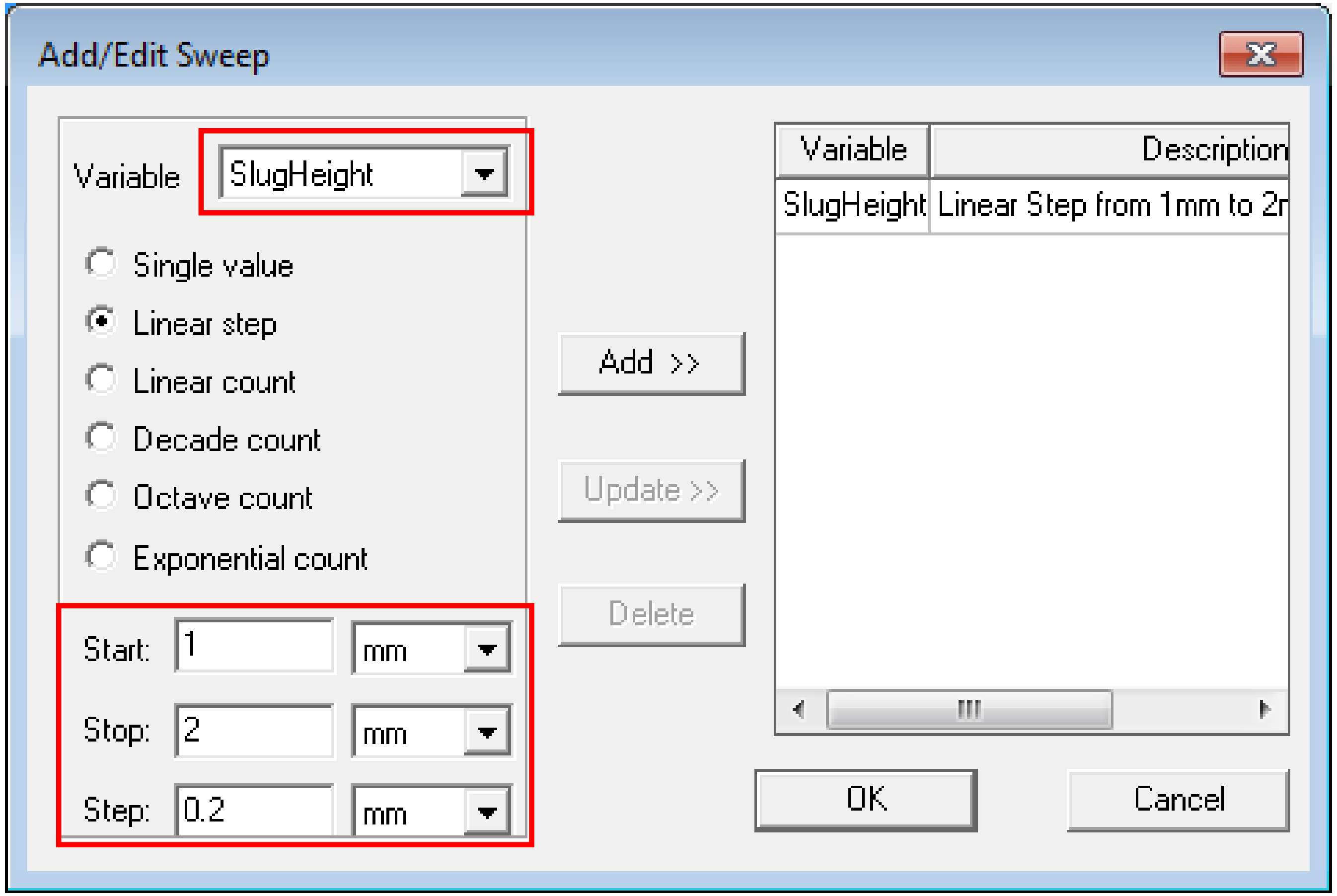
- Plot Mag_B
  - In history tree select the plane **Global:YZ**
  - Menu item ***Maxwell 3D→ Fields →B→ Mag_B***
  - In Create Field Plot window,
    - Press Done



  - **The results show about 0.2 Tesla field in the Center of the Slug which is well within the linear region of the BH curve for most steels**

Parametric Sweep Setup

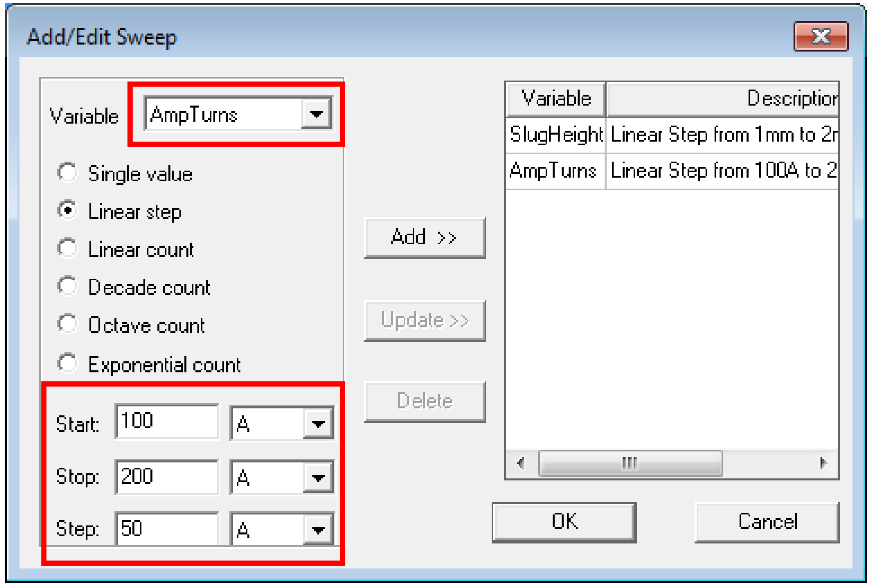
- Launch Setup Sweep Analysis window,
  - Select the menu item *Maxwell 3D → Optimetrics Analysis → Add Parametric*
- Add Parameter Sweep for SlugHeight
  - In Setup Sweep Analysis window, select Add
  - In Add/Edit Sweep window,
    - Variable: SlugHeight
    - Linear Step: ☑ Checked
    - Start: 1mm
    - Stop: 2mm
    - Step: 0.2mm
    - Select Add

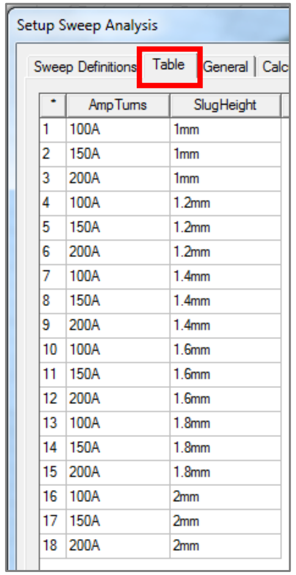


Note: Do not close the Add/Edit Sweep window

Parametric Sweep Setup

- Add Parameter Sweep for **AmpTurns**
  - In Add/Edit Sweep window,
    - Change Variable to **AmpTurns**
    - Linear Step: ☑ **Checked**
    - Start: 100 A
    - Stop: 200 A
    - Step: 50 A
    - Select Add
    - Press OK to close Add/Edit Sweep window
- View Added Parametric Variations
  - In Setup Sweep Analysis window,
  - Change tab to Table to inspect the combination of created solutions. There should be 18 solutions since we defined 6 variations of **SlugHeight** and 3 variations of **AmpTurns**

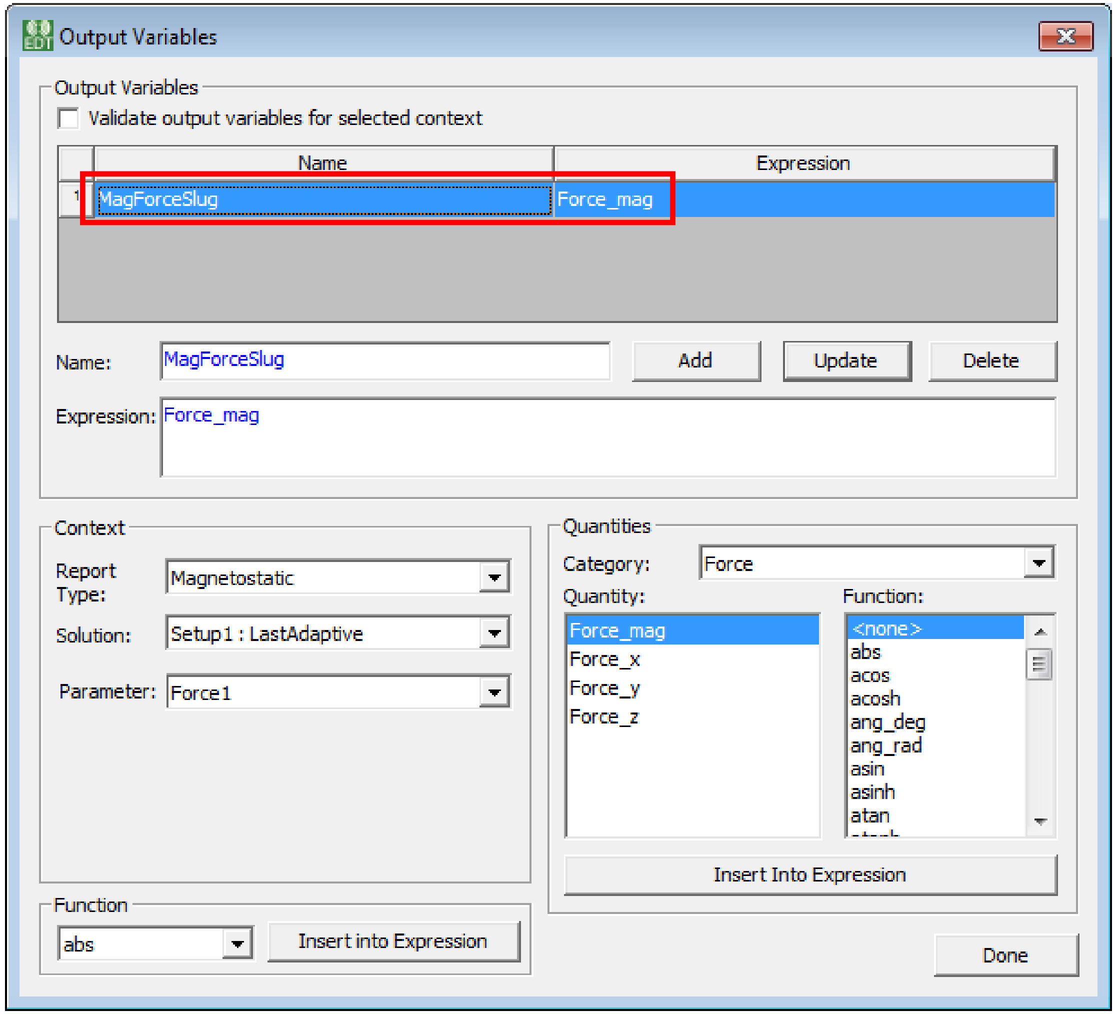




Note: Do not close the Setup Sweep Analysis window

Parametric Sweep Setup

- **Setup Output Parameters: Create Output Variable**
  - **In Setup Sweep Analysis window,**
  - **On Calculations tab, select Setup Calculations**
  - **In Add/Edit Calculation window, Select Output Variables**
  - **In Output Variables window,**
    - **Parameter: Force1**
    - **Category: Force**
    - **Quantity: Force_mag**
    - **Select Insert Into Expression**
    - **Name: MagForceSlug**
    - **Press Add button**
    - **Press Done button**

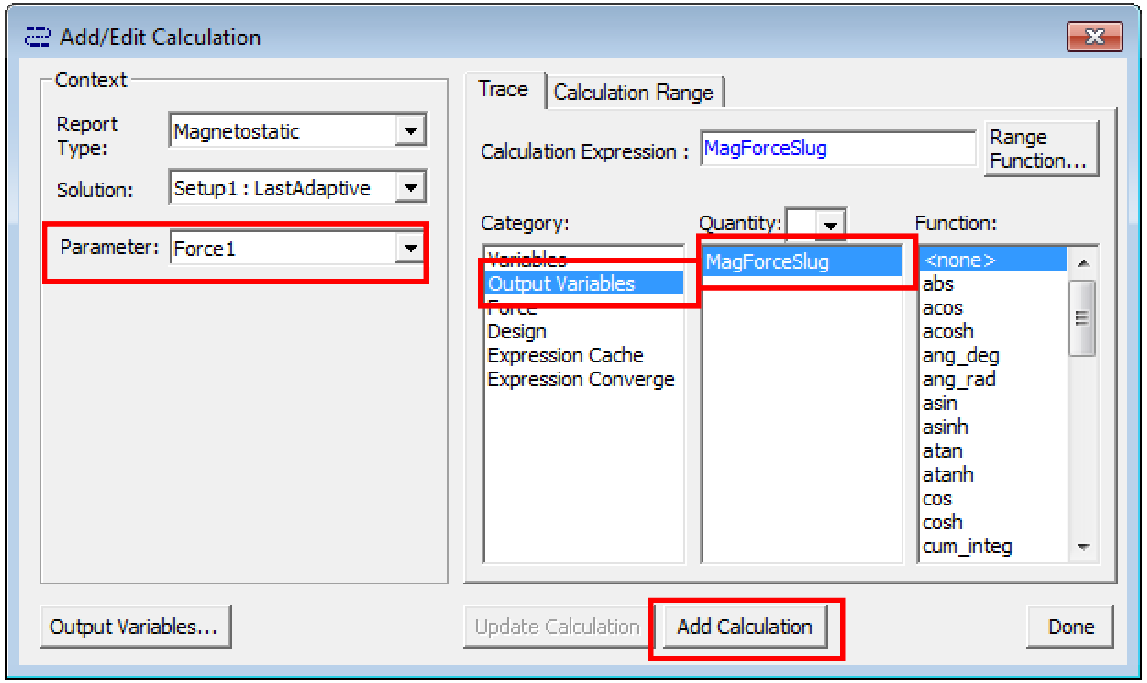


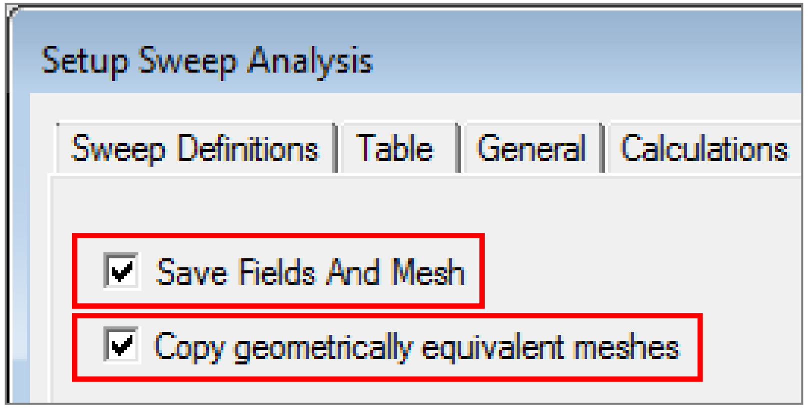
Note: You will now return to Add/Edit Calculations window. Do not close this window

Parametric Sweep Setup

- Setup Output Parameters: Add Output Variable
  - In Add/Edit Calculation window,
    - Parameter: Force1
    - Category: Output Variables
    - Quantity: MagForceSlug
    - Select Add Calculation
    - Select Done to close window

- Setup Options
  - In Setup Sweep Analysis window,
    - Select Options tab on Setup Sweep Analysis window
    - Save Fields and Mesh: ☑ Checked
    - Copy geometrically equivalent meshes: ☑ Checked
    - Click OK to close

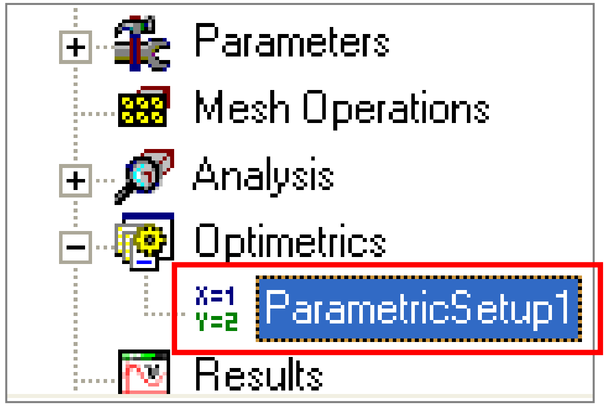




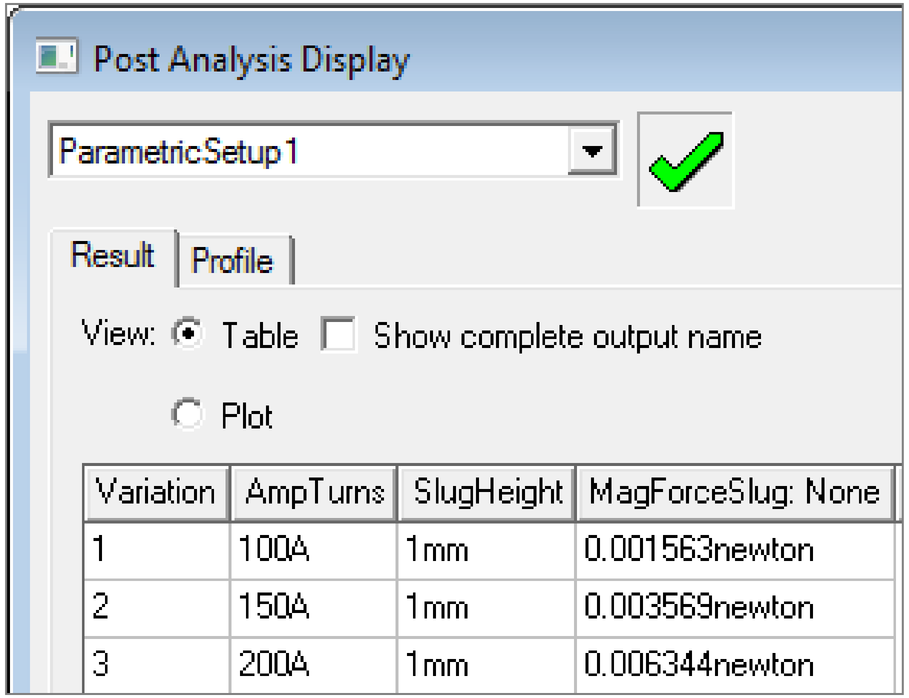
Solve the Parametric Setup

- **Solve Parametric Setup**
  - **In Project manager window, expand the tree for Optimetrics**
  - **In Project manager window** ***RMB on Parametric Setup1 → Analyze***

Note: the solving criteria is taken from the nominal problem Setup1. Each parametric solution will re-mesh if the geometry has changed or the energy error criteria is not met as defined in Setup1



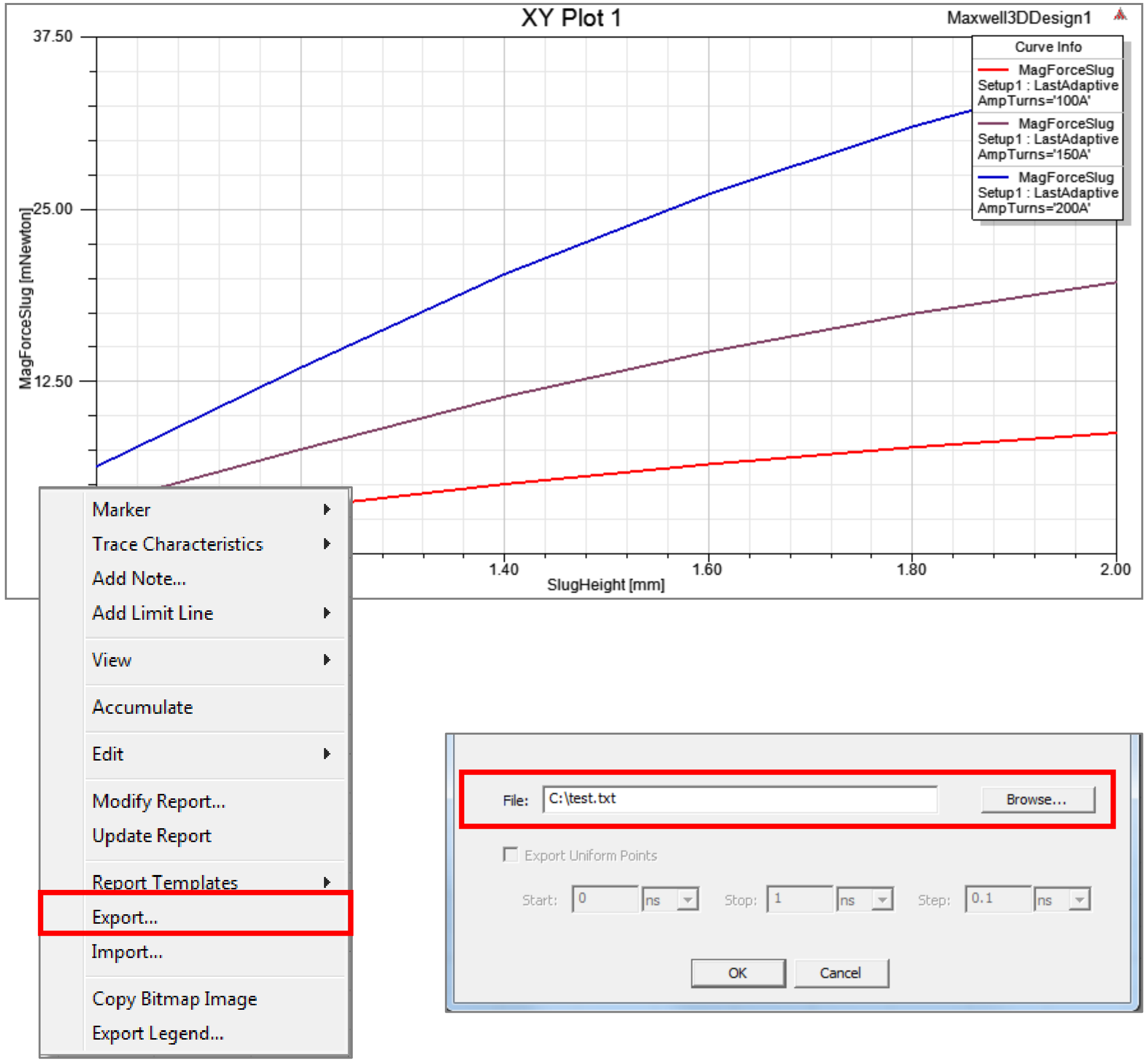
- **View the Results of Parametric Sweep**
  - **In Project manager window, expand the tree for Optimetrics**
  - **In Project manager window** ***RMB on Parametric Setup1 → View Analysis Results***
  - **In Post Analysis Display window,**
    - **View: Select Table to view results in tabular form**



Ansys

Create XY Plot

- **Plot Force vs. AmpTurns vs. SlugHeight**
  - ***Maxwell 3D → Results → Create Magnetostatic Report → Rectangular Plot***
  - **In Report window,**
    - **Parameter: Force1**
    - **Primary Sweep: SlugHeight**
    - **X : Default**
    - **Category: Output Variables**
    - **Quantity: MagForceSlug**
    - **Select the Families tab:**
    - **Ensure that that AmpTurns is selected as the Sweeps variable**
    - **Press New Report**
  - **The Plot should appear as shown in image**
  - **Right click on the plot and select export to export the data in text file format**



Create Data Table

- Create a Table of SlugHeight, Force, and AmpTurns
  - Menu item *Maxwell 3D→ Results→ Create Magnetostatic Report → Data Table*
  - In Report window,
    - Parameter: Force1
    - Parametric Sweep: SlugHeight
    - X : Default
    - Category: Output Variables
    - Quantity: MagForceSlug
    - Select New Report

	SlugHeight [mm]	MagForceSlug [mNewton] Setup1 : LastAdaptive AmpTurns='100A'	MagForceSlug [mNewton] Setup1 : LastAdaptive AmpTurns='150A'	MagForceSlug [mNewton] Setup1 : LastAdaptive AmpTurns='200A'
1	1.000000	1.563387	3.568732	6.344413
2	1.200000	3.396098	7.654352	13.607737
3	1.400000	5.075690	11.385471	20.240827
4	1.600000	6.509813	14.678364	26.094867
5	1.800000	7.775657	17.384163	30.905173
6	2.000000	8.797639	19.749854	35.110848

Saving the Project

- This completes the workshop
- Save the file with the name Workshop_4_2 in the working folder

End of Presentation